higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING FABRICATION - BOILERMAKING
NQF LEVEL 3

(6031013)

21 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 10 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Use only blue or black ink.
5. Write neatly and legibly.

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QUESTION 1

Various options are given as possible answers to the following questions. Choose the correct answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 Welding symbols' parts consist of certain elements like the arrow. A welding symbol can also indicate certain process information.

Which ONE of the following types of *process information* can be indicated by a welding symbol?

A Reference line
B Tail
C Supplementary symbols
D Joint type

1.2 …. is a non-ferrous metal.

A Copper
B Wrought iron
C Mild steel
D Cast iron

1.3 Which marking-off tool may be used to draw a line parallel to the edge of a plate, without the support of any other tool?

A Prick punch
B Odd leg caliper/Jenny caliper
C Divider
D Square

1.4 The point angle related to a center punch is …

A 45°.
B 30°.
C 90°.
D 60°.

1.5 ONE component of the combination set is the …

A center head.
B engineer square.
C tape measure.
D scriber.

1.6 Identify the marking media that will be applied to rough steel structures.

A Copper sulphate
B White wash/chalk
C Marking blue
D Engineer's blue spray

1.7 A gusset's purpose is to …

A connect and strengthen structural steel members.
B carry heavy loads over a long span.
C support overhead floors.
D hold components together to be welded.

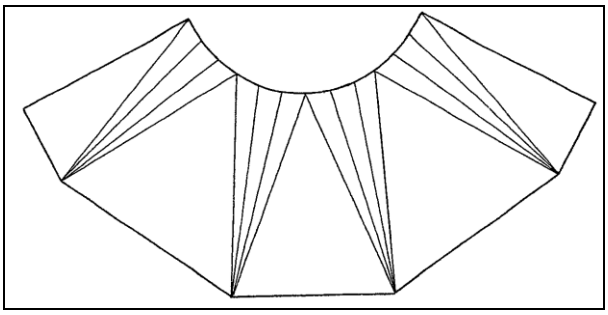
1.8 The correct abbreviation for Gas Metal Arc Welding is …

A GTAW.
B AW.
C GMAW.
D GAW.

1.9 The following formula represents Pythagoras' theorem:

A $A = \pi r^2$
B $V = l \times b \times h$
C $a = \sqrt{b^2 + c^2}$
D $C = \pi D^2$

1.10 The following drawing illustrates a certain drawing layout method. Identify the layout method.



A Parallel line method
B Tri-angulation method
C Radial line method
D First angle orthographic projection

(10 × 1) **[10]**

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write 'True' or 'False' next to the question number (2.1–2.4) in the ANSWER BOOK. Also give ONE explanation for each answer.

2.1 Structural steel sections are formed by means of the following two processes: hot rolled and cold-formed processes.

2.2 A water cooler must never be connected to a GMAW system.

2.3 An overhead traveling crane can be operated by means of a remote control.

2.4 A square-to-round pipe fitting is classified as a transition piece.

(2 × 4) **[8]**

QUESTION 3

3.1 FIGURE 1 indicates a weld symbol with indications of specific welding information. Write the answer next to the question number (3.1.1–3.1.5) and the meaning of the welding information it represents in the ANSWER BOOK.

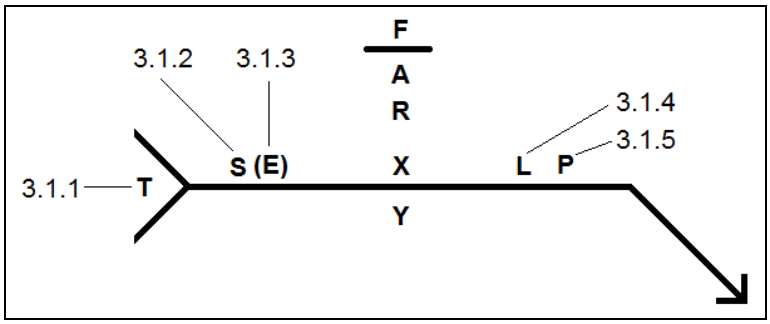


FIGURE 1 (5 × 1) (5)

3.2 Draw a neat drawing of a weld symbol and add the following *basic* and *supplementary* welding symbols to the drawing of the weld symbol. Ensure that the symbols are placed in the correct positions. Write the question numbers (3.2.1–3.2.3) next to the indications of the welding symbols in the ANSWER BOOK.

3.2.1 Fillet weld (TS)

3.2.2 Field weld

3.2.3 Concave finishing

(2 × 3) (6)
[11]

QUESTION 4

4.1 Draw neat drawings illustrating the following butt-joints. Also indicate the position of the bead.

4.1.1 Double-bevel groove weld

4.1.2 Double-V groove weld

(2 × 2) (4)

4.2 Draw neat drawings illustrating the following corner joints. Also indicate the position of the bead.

4.2.1 Inside corner – fillet weld

4.2.2 Outside corner – single bevel groove weld.

(2 × 2) (4)

4.3 Explain the reason why each type of shielding gas must use its own flow meter. (2)

4.4 Differentiate between the *tilt angle* and *slope angle*. (2 × 2) (4)

4.5 List THREE advantages of plasma arc welding. (3 × 2) (6)

4.6 Explain ONE negative result the following welding defects have on a weld:

4.6 4.6.1 Spatter

4.6.2 Undercut

4.6.3 Cracking

(2 × 3) (6)

4.7 State the type of injury that might happen if a welding helmet is not used properly during welding. (1)

[27]

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QUESTION 5

5.1 FIGURE 2 indicates a round tank, fabricated from 12 mm hot rolled sheet. The tank is sealed with a top lid and a bottom base. Take note that the measurements indicated are outside measurements.

Calculate the volume of the tank in m^3 unit. Show all calculations. Answers must be rounded off to three decimals.

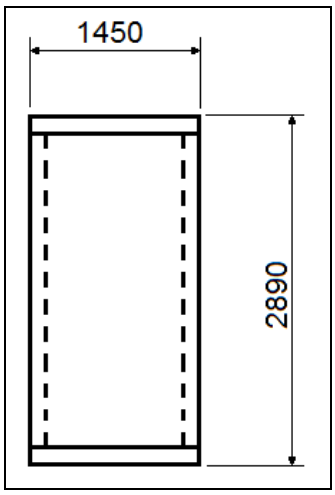


FIGURE 2 (7)

5.3 FIGURE 3 illustrates a specific reading on a vernier calliper. Study the sketch and identify the reading as illustrated.

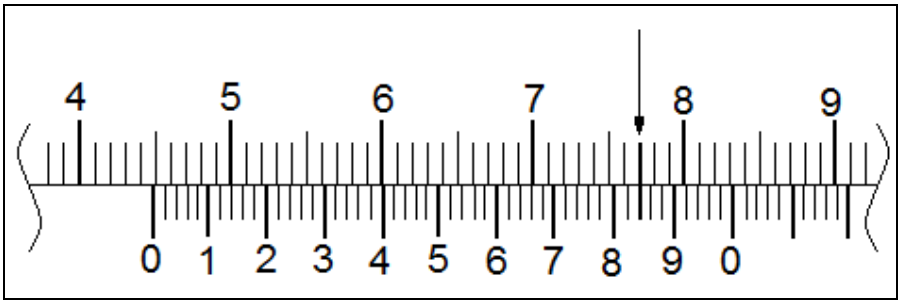


FIGURE 3 (3)

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5.3 FIGURE 4 illustrates a pitched roof truss.

Calculate the length of the rafter, given that the kingpost height is 2 340 mm and the run length is 4 320 mm. Make use of Pythagoras' theorem to do the calculation. Round the answer off to THREE decimals and use meter unit.

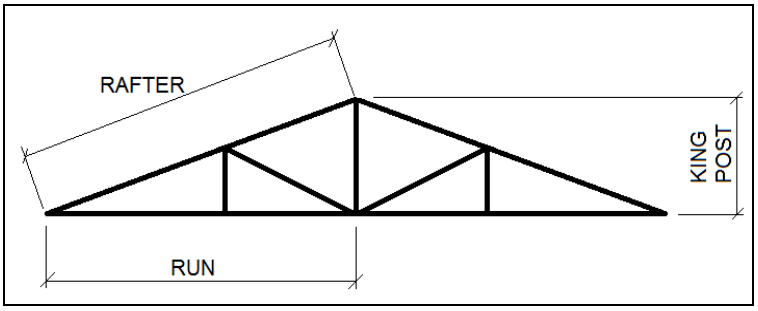


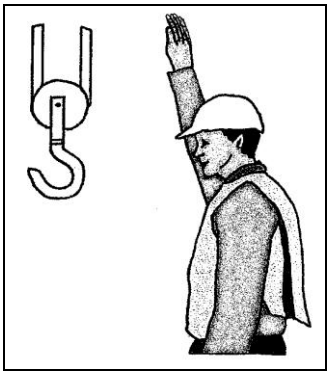
FIGURE 4 (4)

5.4 Which type of crane will normally be used in a fabricating workshop? (1)

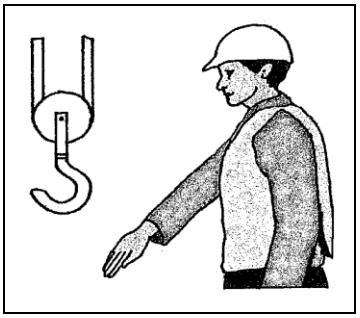
5.5 State TWO advantages of an overhead electric travelling crane that is operated by a remote control. (2)

5.6 The following pictures indicate TWO different hand signals used during rigging work. Write only the name and the meaning of the hand signals next to the question numbers (5.6.1–5.6.2) in the ANSWER BOOK.

5.6.1



5.6.2



(2 × 1) (2)

5.7 List THREE typical main components of a pressure vessel. (3)

[22]

QUESTION 6

6.1 FIGURE 5 illustrates a frame, manufactured from 16 mm square bar.

Calculate the total length of material needed to manufacture the frame. Give the answer in millimetres and round off to three decimals. Pay special attention to the two bends.

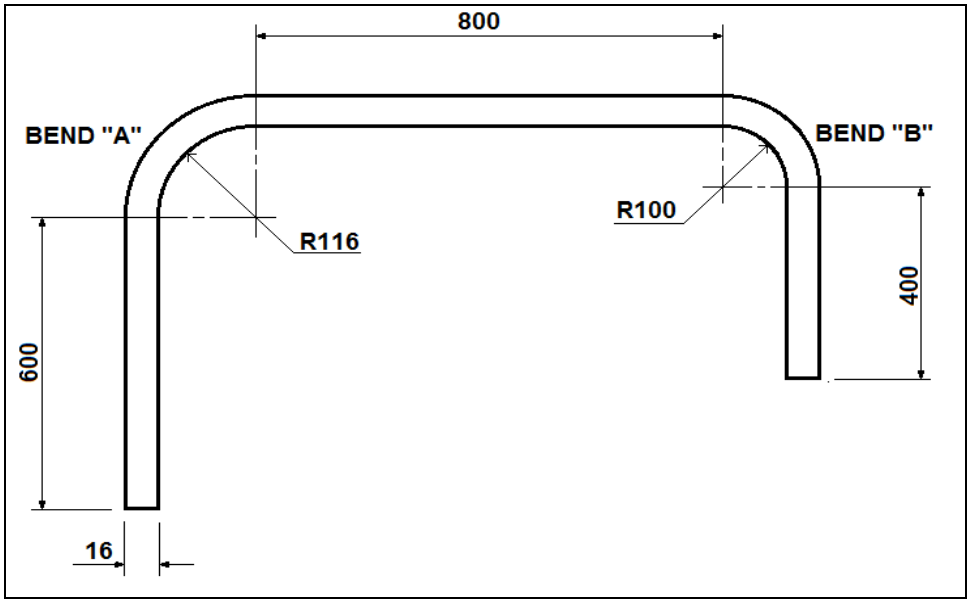


FIGURE 5 (9)

6.2 FIGURE 6 illustrates the front and top views of a square cone.

Re-draw the drawing according to scale 1:1 in your ANSWER BOOK and construct a scale to determine the true lengths of corner line 'OA' and middle seam line 'OX'. The triangulation or radial method can be used.

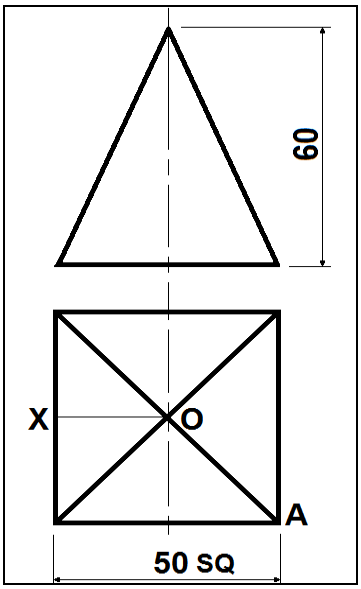


FIGURE 6 (7)

6.3 List THREE factors that have to be considered to ensure the accuracy of calculations during the fabrication process. (3)

6.4 List THREE components of the hand bending brake/pan-and-box folding machine. (3)

[22]

TOTAL: 100

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